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MLS 101: Introduction to Medical Laboratory Science - Specimens, Safety and Accuracy

How laboratory results are produced, and why pre-analytical errors matter

FACULTY PRIMER

Faculty Primer

Faculty of Medical Laboratory Science
Medical Laboratory Science | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Most laboratory errors happen before the sample reaches the analyser. This pack introduces the laboratory disciplines, specimen collection and handling, biosafety, quality control and the meaning of reference ranges, with practical checklists for the teaching laboratory.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe the major laboratory disciplines and the specimen types each handles.
- Explain the pre-analytical, analytical and post-analytical phases of testing.
- Apply biosafety and waste-disposal rules in a teaching laboratory.
- Interpret quality-control results and explain reference ranges.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The laboratory disciplines

Medical laboratory science is a set of specialised diagnostic disciplines, each with its own methods and quality concerns.

- Haematology: blood counts, haemoglobin, coagulation and blood film morphology.
- Clinical chemistry: electrolytes, enzymes, glucose, renal and liver function.
- Medical microbiology: culture, identification and antimicrobial susceptibility testing.
- Histopathology and cytopathology: tissue and cell examination.
- Immunology and serology: antibody and antigen detection.

- Blood transfusion science: grouping, cross-matching and component therapy.

The three phases of testing

A result is only as reliable as the weakest phase in the chain.

- Pre-analytical: patient preparation, collection timing, correct tube, adequate volume, labelling, transport and storage.
- Analytical: calibration, reagent quality, instrument maintenance and technician competence.
- Post-analytical: result validation, reporting, critical-value communication and record retention.
- Haemolysis, delayed transport and under-filled tubes are common causes of rejected samples.
- Label at the bedside, never in advance, and never relabel a specimen.
- Critical results must be phoned to the clinician and documented.

> **Rejection is a safety feature**
> Rejecting a wrongly labelled sample protects the patient from a result belonging to someone else. Never 'fix' a label.

Biosafety and quality control

The laboratory is a controlled environment with rules that exist for a reason.

- Treat all specimens as potentially infectious.
- No eating, drinking, applying cosmetics or storing food in the laboratory.
- Use a biological safety cabinet for procedures that generate aerosols.
- Decontaminate surfaces before and after work; separate clean from dirty zones.
- Sharps into puncture-resistant containers; chemical and biological waste segregated.
- Quality control: run control materials, plot Levey-Jennings charts and stop testing when results fall outside acceptable limits.

The laboratory's place in diagnosis

Laboratory results influence most clinical decisions, which makes accuracy a patient safety issue. The discipline covers haematology, clinical chemistry, microbiology, histopathology, immunology and blood transfusion, each with its own methods and quality requirements.

Every result passes through a chain: patient identification, collection, transport, preparation, analysis, validation and reporting. Most errors occur before the analyser is involved - wrong tube, insufficient volume, delayed transport, mislabelling.

Reference ranges describe the central 95 per cent of a healthy population; they are not thresholds of health. A result outside the range may be normal for that individual, and interpretation always requires clinical context.